

GENERAL MARKING GUIDANCE

- Award one mark for each marking point that is met. A question's marks are the sum of its satisfied marking points and must never exceed the stated maximum.
- Accept any wording listed after "Accept"; the exact phrasing is not required.
- Do not award a marking point that the candidate contradicts elsewhere in the same answer.
- Ignore spelling errors where the intended term is unambiguous.
- Where no answer has been given, award 0 marks.

Question 1 (2 marks)

1. Names the mitochondrion. Accept "mitochondria". (1 mark)
2. States that ATP is produced. Accept "energy in the form of ATP". Do not accept "energy" on its own. (1 mark)

Question 2a (1 mark)

1. glucose + oxygen → carbon dioxide + water. Accept the reactants and the products in either order. Accept a correct symbol equation. (1 mark)

Question 2b (2 marks)

1. Muscle cells have a high energy demand / contract frequently. (1 mark)
2. Links that demand to aerobic respiration in the mitochondria releasing more ATP. (1 mark)
Do not award point 2 for a general statement about needing energy; the answer must refer to respiration or to ATP release.

Question 3 (3 marks) - calculation, method marks apply

1. Both measurements converted to the same unit, e.g. 50 micrometres given as 0.05 mm. (1 mark)
2. Correct method shown: magnification = size of image / size of real object, with the candidate's own figures substituted. (1 mark)
3. Correct answer: x2000. (1 mark)

Partial credit: award marking points 1 and 2 for correct conversion and correct method even when the final value is wrong (error carried forward). Award point 3 only for x2000. A bare correct answer of x2000 with no working scores all 3 marks.

Question 4 (2 marks) - award any two of the following, maximum 2

1. A long, thin extension increases the surface area for uptake. Accept "sticks out into the soil so there is a bigger surface area". (1 mark)
2. Many mitochondria release the ATP needed for active transport of mineral ions. Accept "activeuptake" for active transport, and accept "energy" here. (1 mark)
3. Thin cell wall gives a short diffusion distance. (1 mark)

Question 5 (1 mark)

1. Any one of: controls what enters and leaves the cell; acts as a partially permeable barrier; separates the cell contents from the surroundings. (1 mark)

Question 6 (4 marks)

1. Large surface area, from the very large number of alveoli. (1 mark)
2. The alveolus wall is one cell thick, giving a short diffusion path. (1 mark)
Do not award this point where the candidate describes the wall as thick or muscular.
3. A dense capillary network / good blood supply maintains the concentration gradient. (1 mark)
4. The surface is moist, so gases dissolve before diffusing. (1 mark)

Question 7 (3 marks)

1. The movement of water. Do not accept "movement of particles". (1 mark)
2. From a dilute solution (high water potential) to a concentrated solution (low water potential). Accept the equivalent statement made in terms of solute concentration. (1 mark)
3. Through a partially permeable membrane. Accept "semi-permeable". (1 mark)

Question 8 (2 marks)

1. Lactic acid. (1 mark)
2. Any one effect: muscle fatigue; the muscle stops contracting efficiently; pain or cramp. (1 mark)

Answers continued in the extra answer space are marked in full.

TOTAL: 20 marks